

VIRTUAL JOB SIMULATION REPORT
ON
DATA VISUALISATION: EMPOWERING BUSINESS
WITH EFFECTIVE INSIGHTS

Based on the TATA Forge Virtual Job Simulation

Submitted in partial fulfilment of the academic requirements for the degree of

B.Tech CSE(Data Science)

COMPUTER SCIENCE & ENGINEERING (DATA SCIENCE)



Submitted By

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Roll No.: 25SCS1003002980

Academic Batch: 2025–2029

IILM UNIVERSITY, GREATER NOIDA, U.P.

Completion Date: 26 August 2026

Mode: Self-Paced Virtual Job Simulation

CANDIDATE'S DECLARATION

I, VIDIT RAJ SINGH, hereby declare that the report titled “Data Visualisation: Empowering Business with Effective Insights” has been prepared on the basis of the TATA Forge Virtual Job Simulation completed by me. The report presents the role-based tasks, learning outcomes and practical understanding gained during the simulation.

This was a virtual job simulation and not an internship or employment engagement. The work described in this report is based on the simulation activities completed as part of the program, including business scenario framing, data exploration and analysis, effective visualisation and communication of insights.

I further declare that this report has been prepared for academic presentation purposes and that the information has been organised from the job simulation certificate and the completed presentation material related to the program.

Student Name: Vidit Raj Singh

Roll No.: 25SCS1003002980

Signature: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the TATA Forge Virtual Job Simulation and the Forge platform for providing an opportunity to explore the practical application of data visualisation and business analytics through a structured virtual experience. The simulation helped me understand how data can be interpreted and transformed into meaningful insights for business decision-making.

I am thankful to IILM University, Greater Noida, and the faculty members of the Computer Science and Engineering department for their academic guidance and encouragement. I also acknowledge the value of simulation-based learning, which helps students connect theoretical concepts with realistic professional tasks.

Finally, I would like to thank my family and well-wishers for their constant support and encouragement throughout my academic journey.

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3. JOB SIMULATION CERTIFICATE

3.1 Certificate of Completion

The job simulation was completed in August 2026. The completion certificate records the practical areas covered during the program: Framing the Business Scenario, Choosing the Right Visuals, Creating Effective Visuals, and Communicating Insights and Analysis. The certificate identifies the program as “Data Visualisation: Empowering Business with Effective Insights.”

The certificate is included as supporting evidence for the academic report. The simulation was completed through a virtual, self-paced format and focused on practical role-based tasks related to business analysis and data visualisation.

JOB SIMULATION COMPLETION CERTIFICATE



Inspiring and
empowering
future professionals

Vidit Raj Singh

Data Visualisation: Empowering Business with Effective Insights

Certificate of Completion
August 26th, 2026

Over the period of August 2026, Vidit Raj Singh has completed practical tasks in:

Framing the Business Scenario
Choosing the Right Visuals
Creating Effective Visuals
Communicating Insights and Analysis

Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6a8b2b6509f1e253b84b87ba | User Verification Code 6a8b2a88806df61d2fbffe96 | Issued by Forage

4. JOB SIMULATION AND ORGANIZATION OVERVIEW

4.1 Introduction

Data visualisation has become an important part of modern business analysis because organisations generate large volumes of information that must be understood and communicated effectively. Raw data may contain useful patterns, trends and relationships, but these insights can remain hidden unless the information is analysed and presented in an understandable form.

The virtual job simulation, titled “Data Visualisation: Empowering Business with Effective Insights,” provided a structured learning journey focused on the relationship between data, business context and communication. The simulation was organised around four sequential tasks: framing the business scenario, exploring and analysing data, creating effective visualisations, and presenting insights to leadership.

The purpose of the simulation was not simply to create charts or dashboards. It also emphasised the importance of asking the right business questions, identifying meaningful patterns and communicating findings in a form that can support decision-making.

4.2 About the Organization and Simulation Platform

TATA is presented in the simulation material as a global conglomerate associated with innovation, engineering excellence and digital transformation across industries. The job simulation experience was designed to provide exposure to business-oriented analytical thinking and professional workflows in a virtual learning environment.

The simulation was completed through the Forge/Forage virtual experience platform. The platform provides structured, role-based simulations that allow participants to engage with practical tasks and develop industry-relevant skills through guided activities.

4.3 Job Simulation Overview

Candidate: Vidit Raj Singh

University: IILM University, Greater Noida

Degree: B.Tech CSE (Data Science)

Program: TATA Forge Virtual Job Simulation

Domain: Data Visualisation & Business Analytics

Mode: Self-paced

Duration: Approximately 3–4 hours

Completion Date: 26 August 2026

4.4 Industry Challenge

A major challenge in data-driven organisations is that large amounts of information do not automatically result in better decisions. Decision-makers need information that is

relevant, accurate, understandable and connected to business objectives. Poorly selected charts, unnecessary complexity or weak communication can make important findings difficult to interpret.

The industry challenge therefore involves more than analysing a dataset. A data professional must understand the business situation, identify relevant questions, explore available information and select visual forms that clearly communicate the findings.

5. OBJECTIVES AND SCOPE

5.1 Core Objectives

The first objective was to develop an understanding of business context before beginning detailed data analysis. The simulation emphasised that useful analysis begins with the right questions and a clear understanding of what decision-makers need to know.

The second objective was to strengthen data analysis and interpretation skills through exploring datasets, identifying patterns and trends, and distinguishing useful insights from unnecessary information.

The third objective was to understand effective visualisation design. Data should be presented using visuals that match the analytical purpose and make the intended message easier to understand.

The fourth objective was to improve business storytelling and professional communication so that findings and recommendations can be communicated clearly to stakeholders and decision-makers.

5.2 Solution Approach and Scope

The simulation approached the industry challenge through a sequential workflow. First, the business scenario was framed to identify context and relevant questions. Next, data was explored and analysed to uncover patterns and observations. These findings were translated into appropriate visual representations and then communicated to a leadership-oriented audience.

The scope covered business problem understanding, data exploration, selection of suitable visualisations and presentation of insights. The experience demonstrated how analytical and communication skills must work together.

5.3 Skills and Tools

The simulation focused on data analysis, interpretation, visualisation design, business storytelling and professional communication. The simulation material also referenced industry-standard visualisation tools such as Power BI or Tableau for creating interactive dashboards and visual narratives.

6. TASK-WISE LEARNING AND SIMULATION EXPERIENCE

6.1 Task 1 – Framing the Business Scenario

The first task focused on understanding the business scenario before beginning detailed analysis. The purpose was to learn how to think from the perspective of a business leader and prepare relevant strategic questions for senior executives.

This stage demonstrated that data analysis should not begin with random exploration. Analysts need to understand the business context and identify what information is required to support a particular decision. A clearly framed question helps guide later stages of analysis and visualisation.

6.2 Task 2 – Data Exploration and Analysis

The second task focused on exploring data to identify useful patterns, trends and insights. The simulation journey described this stage as involving the cleaning, analysis and examination of complex datasets.

The purpose of data exploration is to move from raw information towards a clearer understanding of the underlying business situation. At this stage, an analyst looks for relationships and observations that may help answer the business questions identified earlier.

6.3 Task 3 – Building Effective Visualisations

The third task focused on translating analytical findings into effective visual representations. The simulation material described the use of dashboards and compelling visual narratives through tools such as Power BI or Tableau.

An effective visualisation should make important information easier to understand. The selection of a chart or dashboard should depend on the message being communicated and the audience receiving the information.

6.4 Task 4 – Presenting Insights to Leadership

The final task focused on communicating findings and recommendations to a leadership audience. This stage connected the earlier analytical work with business communication and decision-making.

The objective was to present insights in a clear, concise and meaningful manner. Leadership communication requires an analyst to explain not only what the data shows but also why the finding matters and what action or consideration may follow from it.

7. LEARNING OUTCOMES AND CONCLUSION

7.1 Domain and Technical Learning

The simulation provided practical understanding of the data visualisation workflow. The main learning areas included business scenario framing, data exploration, interpretation of patterns, selection of appropriate visuals and communication of analytical findings.

The experience also strengthened awareness of the relationship between data analytics and business intelligence. Technical analysis becomes more valuable when results are transformed into understandable information that can support real decisions.

7.2 Professional Learning

The simulation developed professional skills related to strategic thinking, analytical reasoning, communication and structured problem-solving. It demonstrated the importance of considering the audience when preparing analytical outputs and communicating recommendations.

7.3 Key Challenges and Takeaways

The major conceptual challenge addressed by the simulation was moving from raw data to a meaningful business narrative. This requires understanding the business need, identifying relevant information, exploring data carefully, selecting effective visuals and communicating the final insight appropriately.

A key takeaway was that data visualisation is not limited to chart creation. It is part of a larger process that connects data with context, interpretation and action.

7.4 Conclusion

The TATA Forge Virtual Job Simulation on “Data Visualisation: Empowering Business with Effective Insights” provided a structured role-based learning experience in data visualisation and business analytics. The four tasks represented a complete progression from understanding a business scenario to communicating insights to leadership.

Overall, the experience contributed to my development as a B.Tech Computer Science and Engineering (Data Science) student by providing practical exposure to analytical thinking, data storytelling and professional communication.

8. REFERENCES

1. Certificate of Completion, “Data Visualisation: Empowering Business with Effective Insights,” issued through Forage, August 2026.
2. TATA Forge Virtual Job Simulation presentation material prepared for the completed job simulation.
3. Simulation Task 1: Framing the Business Scenario.
4. Simulation Task 2: Data Exploration and Analysis.
5. Simulation Task 3: Building Effective Visualisations.
6. Simulation Task 4: Presenting Insights to Leadership.

END OF REPORT